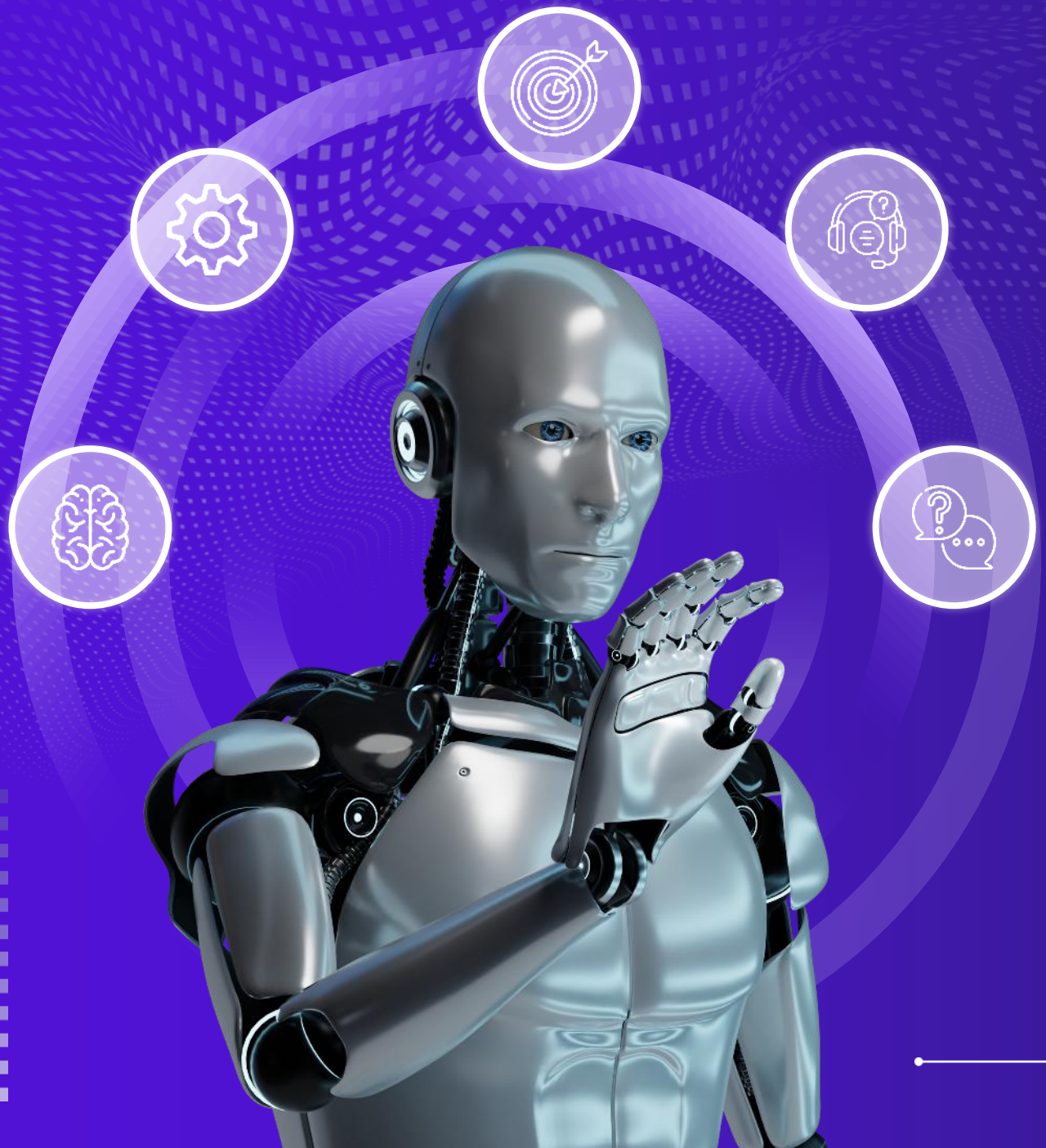

Agentic AI

Agentic AI on IBM Cloud

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What is Agentic AI

Agentic AI integrates **large language models (LLMs), machine learning, and enterprise automation** to create autonomous AI agents capable of **decision-making, goal-setting, problem-solving, and adapting to unstructured environments**, unlike rule-based AI.

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Evolution of Agentic AI

Early AI Models
Focused on predefined rules and automation.

Machine Learning Era
AI began adapting and learning patterns.

Rise of Large Language Models (LLMs)
Enhanced contextual understanding.

Agentic AI Integration
Combining AI, automation, and real-time adaptability.

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How Agentic AI Works

Understanding

AI processes data using LLMs and machine learning.

Goal-Setting

Identifies objectives and plans strategies.

Execution

Automates tasks with minimal human input.

Learning

Continuously improves through interactions.

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Benefits Of Agentic AI

Efficiency

Reduces manual work and automates complex tasks, improving productivity.

Scalability

AI agents can adapt to business growth and evolving demands.

Adaptability

Responds dynamically to real-time changes and unstructured environments.

Collaboration

Works alongside humans, enhancing decision-making and workflow efficiency.

Cost Reduction

Lowers operational expenses by automating repetitive and resource-intensive processes.

Continuous Learning

Improves performance over time through data-driven insights and machine learning.

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Agentic AI Vs. Generative AI

Agentic AI

- Autonomous
- Goal-Oriented
- Real-time Environment
- Utilize Tools and APIs
- Continuous Adaptation
- Autonomous Systems
- Independent Decision-Making

Vs

Generative AI

- Creative
- Output-Oriented
- Content-Based
- No external tools
- No adaptation from real-time
- Content Creation
- Make decision based on input patterns

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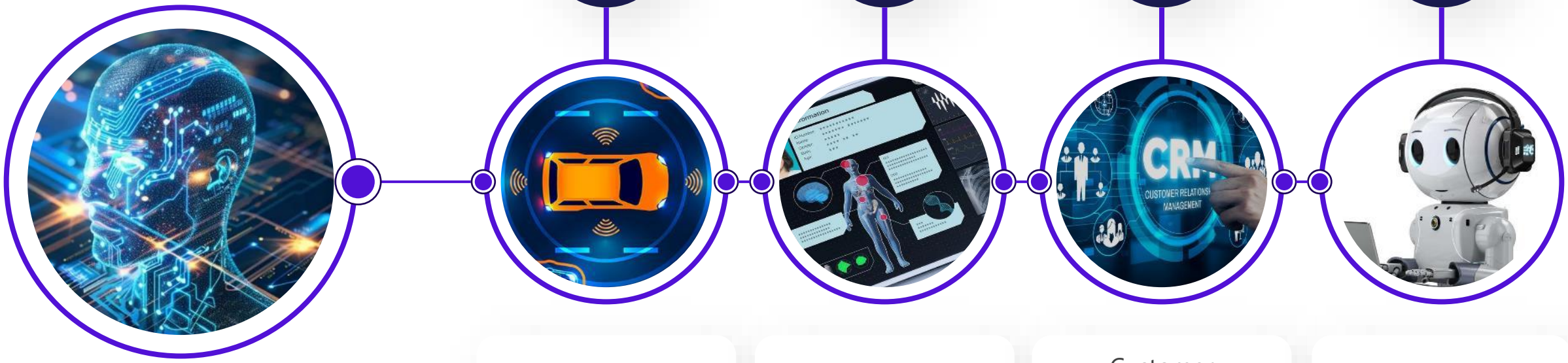
Applications Of Agentic AI



Autonomous Vehicles

Advanced driver assistance systems

Healthcare

Electronic health records

Finance

Customer relationship management systems

Customer Service

Sentiment analysis chatbot

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Use Cases of Agentic AI & Automation

Automated Data Analysis
AI interprets trends in real-time.

Supply Chain Optimization
Predictive analytics improve logistics.

IT & Security
AI-driven cybersecurity threat detection.

Self-Healing Systems
AI resolves technical issues autonomously.

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Risks & Challenges of Agentic AI

Risks

- ✓ Bias & Fairness Issues
- ✓ Security Threats
- ✓ Loss of Human Control

Challenges

- ✓ Ethical Concerns
- ✓ Regulatory & Compliance Issues
- ✓ Accountability & Transparency

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Best Practices For Implementing

01

Ethical AI Development

Ensure transparency and fairness in decision-making.

02

Robust Security Measures

Prevent unauthorized access and cyber threats.

03

Continuous Learning

Enable AI to adapt and improve with new data.

04

Human Oversight

Maintain control to align AI decisions with business values.

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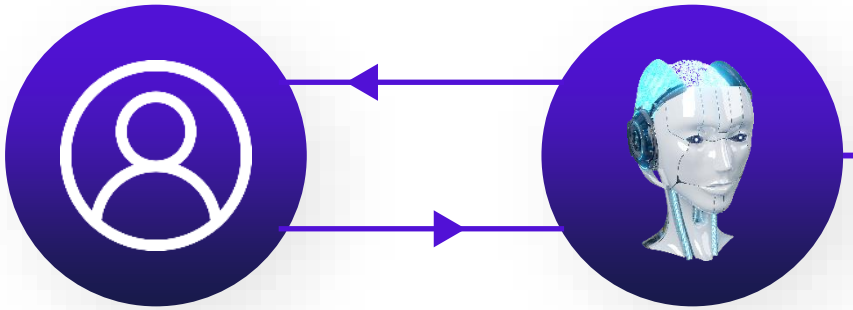
Agentic AI System



User Request

AI Agent

Perception

Multi-modal fusion

- Camera
- Text
- Audio
- Sensors

Cognition

- Memory
- Knowledge Base

Decision making

Action

Executing Tasks

- Physical actions in real-world
- Monitors

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Comparison: LLaMA vs Mistral Large

Model	Provider	Size	Modality	Strengths	Agent Support
LLaMA 3 70B Instruct	Meta	70B	Text	Strong reasoning & instruction-following	Good
LLaMA 3 11B Vision	Meta	11B	Text + Vision	Multimodal capabilities	Moderate
LLaMA 3.3 70B FP8	Meta	70B (FP8)	Text	Optimized for inference	Good
Mistral Large	Mistral AI	40B+ (est.)	Text	Advanced reasoning, native JSON	Strong

Conclusion & Key Takeaways

Agentic AI is revolutionizing **automation beyond simple processes**. Businesses must adapt as AI shapes the future of work. Ethical **AI development** is essential for responsible use, and AI will continue evolving, driving innovation and transformation.

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